

RoadPulse ERD v3 — Traffic · Incident · ASOS 통합 구조

WEATHER STATIONS 기상청 ASOS 관측소		
PK	weather_station_id	VARCHAR(10)
	station_name	VARCHAR(100)
	latitude	DECIMAL(10,7)
	longitude	DECIMAL(10,7)
	elevation_m	DECIMAL(8,2)
	is_active	BOOLEAN

WEATHER MEASUREMENTS 기상청 ASOS 시간자료		
PK	id	BIGINT
FK	weather_station_id	VARCHAR(10)
	observed_at	DATETIME
	temperature_c	DECIMAL(5,2), NULL
	rainfall_mm	DECIMAL(7,2), NULL
	humidity_pct	DECIMAL(5,2), NULL
	wind_speed_ms	DECIMAL(6,2), NULL
	wind_direction_deg	SMALLINT, NULL
	pressure_hpa	DECIMAL(7,2), NULL
	collected_at	DATETIME
UQ: weather_station_id + observed_at		

MODEL VERSIONS 학습 모델 및 평가 이력		
PK	model_version	VARCHAR(50)
	algorithm	VARCHAR(50)
	hyperparameters	JSON
	mae	DECIMAL(12,4)
	rmse	DECIMAL(12,4)
	r2_score	DECIMAL(10,6)
	artifact_path	VARCHAR(500)
	trained_at	DATETIME
	is_active	BOOLEAN

TRAFFIC PREDICTIONS 지점별 1시간 후 교통량 예측		
PK	id	BIGINT
FK	spot_id	VARCHAR(20)
FK	model_version	VARCHAR(50)
	predicted_at	DATETIME
	target_at	DATETIME
	predicted_volume	INT
	actual_volume	INT, NULL
UQ: spot_id + target_at + model_version		

TRAFFIC SPOTS 서울시 SpotInfo		
PK	spot_id	VARCHAR(20)
	spot_name	VARCHAR(150)
	tm_x	DECIMAL(12,6)
	tm_y	DECIMAL(12,6)
	latitude	DECIMAL(10,7)
	longitude	DECIMAL(10,7)
	created_at	DATETIME
	updated_at	DATETIME

TRAFFIC VOLUME MEASUREMENTS 서울시 VollInfo · ML 핵심 학습 데이터		
PK	id	BIGINT
FK	spot_id	VARCHAR(20)
	measured_at	DATETIME
	direction_code	TINYINT
	lane_no	SMALLINT
	traffic_volume	INT
	collected_at	DATETIME
UQ: spot_id + measured_at + direction_code + lane_no		

TRAFFIC SPOT ROAD MAPS SpotInfo 좌표 ↔ 링크 Vertex 공간 매핑		
PK/FK	spot_id	VARCHAR(20)
PK/FK	link_id	VARCHAR(20)
	match_distance_m	DECIMAL(8,2)
	match_method	VARCHAR(30)
	is_primary	BOOLEAN
	created_at	DATETIME
PK: spot_id + link_id		

ROAD SEGMENTS 서울시 LinkInfo 기준정보		
PK	link_id	VARCHAR(20)
	road_name	VARCHAR(150)
	start_node_name	VARCHAR(150)
	end_node_name	VARCHAR(150)
	length_m	INT
	region_code	VARCHAR(20)
	created_at	DATETIME
	updated_at	DATETIME

TRAFFIC SPEED MEASUREMENTS 서울시 TrafficInfo		
PK	id	BIGINT
FK	link_id	VARCHAR(20)
	measured_at	DATETIME
	speed_kmh	DECIMAL(6,2)
	travel_time_sec	INT
	collected_at	DATETIME
UQ: link_id + measured_at		

INCIDENTS 서울시 Incident API · AccInfo		
PK	incident_id	VARCHAR(30)
FK	link_id	VARCHAR(20), NULL
	occurred_at	DATETIME
	expected_clear_at	DATETIME, NULL
	incident_type	VARCHAR(20)
	incident_detail_type	VARCHAR(20)
	road_code	VARCHAR(20)
	tm_x / tm_y	DECIMAL(12,6)
	description	TEXT
	collected_at	DATETIME